

2024, Mar. 2024 Release LOC:Acute Adult

Sickle Cell Disease

Utilization Benchmarks: *Multiple options available, see table below for options*

Overview

Select Day

Initial review ^(1, 2)Episode Day 1 ^(1, 30)Episode Day 2 ⁽³³⁾Episode Day 3 ⁽⁴¹⁾

Episode Day 4

Episode Day 5

Discharge Screens ⁽⁴²⁾

Utilization Benchmarks

Notes

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Overview

Instruction: This subset is for patients with sickle cell crisis including those sickle cell patients presenting with fever of unknown origin, acute chest syndrome, acute pain crisis, aplastic crisis, and vaso-occlusive crisis. For patients with priapism or hyphema, use the General Medical subset.

Introduction: Sickle cell disease is an inherited disease of the red blood cells. Although it most commonly affects people of African descent, it can be found in people of any ethnic descent including South American, Mediterranean, Middle Eastern, or Asian Indian. These patients have a single genetic defect resulting in an abnormal amino acid sequence. Consequently, the red blood cells become deformed into a sickle shape following the release of oxygen. These sickle shaped cells become rigid, sticky, and have a short life span ranging from 10 to 20 days. For most patients, sickle cell disease is considered incurable. Bone marrow transplants are becoming a more common practice. The American Society of Hematology is in the process of developing bone marrow transplantation guidelines. The National Heart, Lung and Blood Institute recommends the use of hydroxyurea in patients with sickle cell related pain that interferes with daily activities (Brandow et al., Blood advances 2020, 4: 2656-701). Acute complications warranting hospitalization include acute chest syndrome, aplastic crisis, infection of unknown origin (for sickle cell patients only) and vaso-occlusive crisis.

Acute chest syndrome

Acute chest syndrome (ACS) is the leading cause of death from sickle cell disease and may be caused by infection, fat emboli, atelectasis, or lung infarction, resulting in obstruction of the pulmonary vasculature. However, in most cases a cause cannot be identified. ACS is more commonly diagnosed in children with sickle cell disease but is more severe in adults. Presenting symptoms of acute chest syndrome resemble pneumonia and may include fever, cough, chest pain, or increased work of breathing as evidenced by dyspnea, wheezing, retractions, and rales. A new pulmonary infiltrate on chest radiograph (CXR) is considered a hallmark sign of acute chest syndrome. A new infiltrate may not be seen initially so it is recommended that serial CXRs be included in the diagnostic work-up while respiratory symptoms persist. Pleural effusions, multilobe disease, and persistent hypoxia despite supplemental oxygen therapy are predictive of a severe course (Ballas, Clinical hemorheology and microcirculation 2018, 68: 105-28; Brandow et al., Blood advances 2020, 4: 2656-701).

Treatment for ACS includes adequate ventilation, anti-infective therapy, maintenance of euvolemia, analgesic administration, and antipyretics as needed. Bronchodilators should be considered if there is a history of reactive airway disease or if wheezing is present. Frequent incentive spirometry should be encouraged. Transfusion with packed red blood cells (PRBCs) should be considered for symptomatic disease if the hemoglobin is 1.0 g/dL (10 g/L) less than baseline. In patients with actual or impending respiratory failure or decreased hemoglobin despite transfusion

with PRBCs, exchange transfusion is recommended (Ballas, Clinical hemorheology and microcirculation 2018, 68: 105-28; National Heart Lung and Blood Institute, Evidence-based management of sickle cell disease: Expert panel report. 2014).

Acute pain crisis

The most common cause of acute pain in Sickle cell patients is vaso-occlusive crisis (VOC). VOC is an acute episode in which ischemic tissue injury is caused by lack of blood flow to a localized area. This decreased blood flow is attributed to a sickled erythrocyte obstruction in the providing blood vessel. When this is amplified many times, these episodes produce tissue hypoxia and lead to pain, tissue and organ damage. The crisis can but does not need to be precipitated by infection, fever, hypoxia, sleep apnea, dehydration, or exposure to extreme temperatures. Pain caused by VOC is the most common complication of sickle cell disease.

The diagnosis of VOC is based upon the history and physical examination. Laboratory tests are not useful to confirm the disorder but may be ordered to rule out other diagnoses. Analgesic administration should be individualized based on the patient's home regimen. Aggressive and timely pain management is key to rapid resolution of these painful episodes, and should be initiated within 60 minutes of arrival to medical center. In order to avoid under treatment of pain, patient controlled or scheduled analgesia are recommended for pain management. Additional supportive measures include hydration (oral and/or intravenous) to maintain euvolemia, supplemental oxygen targeting a room air oxygen saturation of 95%(0.95) percent or greater, and incentive spirometry. Patients that present with acute pain should be evaluated for other etiologies. An abdominal ultrasound and liver function tests may be required to rule out appendicitis, cholecystitis, or pancreatitis. If respiratory symptoms are present, such as cough, chest pain, or hypoxia, a chest radiograph should be performed due to the increased risk of acute chest syndrome. For patients with evidence of encephalopathy, renal and liver function should be tested to rule out multisystem organ failure (Brandow et al., Blood advances 2020, 4: 2656-701; Ballas, Clinical hemorheology and microcirculation 2018, 68: 105-28).

Aplastic crisis

Aplastic crisis is characterized by a decrease in either one or both hemoglobin and reticulocyte values. A hemoglobin decrease of greater than or equal to 2g/dL from patients known baseline is considered to be an acute finding. Aplastic crisis is usually brought on by an infection but can also be caused by medications or autoimmune disorders. Aplastic crisis is treated by transfusing packed red blood cells and maintaining euvolemia. Patients with aplastic crisis may require frequent or numerous transfusions and need to be monitored for complications including alloimmunization, iron overload and delayed hemolytic transfusion reactions (Chou et al., Blood advances 2020, 4: 327-55; Ballas, Clinical hemorheology and microcirculation 2018, 68: 105-28; Porter, Emergency Medicine Clinics 2018, 36: 567-76).

Infection of unknown origin

This subset is for the review of patients with an infection in which the source has not yet been identified. This includes criteria for patients with a fever and infectious symptoms which have failed outpatient management. Once a source is identified, patients are to be directed to the appropriate infection specific subset.

Many patients with sickle cell disease are at high risk for readmission, and careful attention to discharge planning is essential.

InterQual® criteria are derived from the systematic, continuous review and critical appraisal of the most current evidence-based literature and include input from our independent panel of clinical experts. To generate the most appropriate recommendations, a comprehensive literature review of the clinical evidence was conducted. Sources searched included PubMed, AHRQ Comparative Effectiveness Reviews, the National Institute for Health and Care Excellence (NICE), the Centers for Medicare and Medicaid Services (CMS), The Cochrane Library, and The Joint Commission. Other medical literature databases, medical content providers, data sources, regulatory body websites, and specialty society resources may also have been utilized. Relevant studies were assessed for risk of bias following principles described in the *Cochrane Handbook*. The resulting evidence was assessed for consistency, directness, precision, effect size, and publication bias. Observational trials were also evaluated for the presence of a dose-response gradient and the likely effect of plausible confounders. The majority of the content in this subset is based on a limited number of key sources (Brandow et al., Blood advances 2020, 4: 2656-701; Chou et al., Blood advances 2020, 4: 327-55; Ballas, Clinical hemorheology and microcirculation 2018, 68: 105-28; Liem et al., Blood advances 2019, 3: 3867-97).

(Excludes PO medications unless noted)

Select Day, One:● **Initial review, One:** ^(1, 2)

(From arrival in ED through decision to admit)

(Symptom or finding within 24h)

● **OBSERVATION, ≥ One:** ⁽³⁾● Acute pain and failed outpatient analgesic and, **Both:** ^(4, 5, 6)– Analgesic 2 to 3 doses within 24h ^(7, 8)– IV fluid ^(9, 10, 11)● Infection, actual or suspected, **Both:**– Temperature 100.4-101.2°F(38.0-38.4°C) ^(12, 13)

– Anti-infective

● **ACUTE, ≥ One:** ⁽¹⁴⁾● Acute chest syndrome, **One:** ⁽¹⁵⁾● New infiltrate confirmed on chest x-ray and, ≥ **One:** ⁽¹⁶⁾

– Chest pain

– O₂ sat < 92%(0.92) and < baseline ^(17, 18)– Temperature ≥ 100.4°F(38.0°C) ⁽¹²⁾● Respiratory symptom, ≥ **One:** ⁽¹⁵⁾

– Dyspnea

– Cough

– Wheeze

– Retractions

– Rales

● Aplastic crisis, ≥ **One:** ^(19, 20)– Hb ≥ 2 g/dL(20 g/L) below baseline ⁽¹⁷⁾

– Hb < 5.5 g/dL(55 g/L)

– Temperature ≥ 101.3°F(38.5°C) and infection, actual or suspected ^(12, 13)● Acute pain and, **Both:** ^(4, 7, 8, 10)– Analgesic ≥ 4x/24h or continuous ^(7, 21)– IV fluid ^(9, 10, 11)● **INTERMEDIATE, ≥ One:** ⁽²²⁾– NIPPV ⁽²³⁾– Oxygen ≥ 40%(0.40) ≤ 2d ⁽²⁴⁾● **CRITICAL, ≥ One:** ⁽²⁵⁾– ECMO or ECLS ⁽²⁶⁾– Exchange transfusion ^(27, 28)– Hemodynamic monitoring, invasive ⁽²⁹⁾

– Mechanical ventilation

● NIPPV and, ≥ **One:** ⁽²³⁾– Arterial Po₂ ≤ 39 mmHg(5.2 kPa)● Arterial or venous Pco₂ ≥ 60 mmHg(8.0 kPa) and, **One:**

– COPD and arterial or venous pH ≤ 7.24

– No hx of COPD

– Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)

● **Episode Day 1, One:** ^(1, 30)

(Symptom or finding within 24h)

● **OBSERVATION, ≥ One:** ⁽³⁾● Acute pain and failed outpatient analgesic and, **Both:** ^(4, 5, 6)– Analgesic 2 to 3 doses within 24h ^(7, 8)– IV fluid ^(9, 10, 11)● Infection, actual or suspected, **Both:** ⁽³¹⁾– Temperature 100.4-101.2°F(38.0-38.4°C) ^(12, 13)

- Anti-infective
- **ACUTE, ≥ One:** ⁽¹⁴⁾
 - Acute chest syndrome, **Both:** ⁽¹⁵⁾
 - New infiltrate confirmed on chest x-ray and, ≥ **One:** ⁽¹⁶⁾
 - Chest pain
 - O₂ sat < 92%(0.92) and < baseline ^(17, 18)
 - Temperature ≥ 100.4°F(38.0°C) ⁽¹²⁾
 - Respiratory symptom, ≥ **One:** ⁽¹⁶⁾
 - Dyspnea
 - Cough
 - Wheeze
 - Retractions
 - Rales
 - Anti-infective ⁽³²⁾
 - Aplastic crisis, **Both:** ⁽¹⁹⁾
 - Finding, ≥ **One:** ⁽²⁰⁾
 - Hb ≥ 2 g/dL(20 g/L) below baseline ⁽¹⁷⁾
 - Hb < 5.5 g/dL(55 g/L)
 - Intervention, **Both:**
 - Hct or Hb monitoring at least 2x/24h
 - IV fluid or blood product transfusion ⁽⁹⁾
 - Temperature ≥ 101.3°F(38.5°C) and anti-infective ^(12, 13)
 - Acute pain and, **Both:** ^(4, 8, 10)
 - Analgesic ≥ 4x/24h or continuous ^(7, 21)
 - IV fluid ^(9, 11)
- **INTERMEDIATE, ≥ One:** ⁽²²⁾
 - NIPPV ⁽²³⁾
 - Oxygen ≥ 40%(0.40) ≤ 2d ⁽²⁴⁾
- **CRITICAL, ≥ One:** ⁽²⁵⁾
 - ECMO or ECLS ⁽²⁶⁾
 - Exchange transfusion ^(27, 28)
 - Hemodynamic monitoring, invasive ⁽²⁹⁾
 - Mechanical ventilation
 - NIPPV and, ≥ **One:** ⁽²³⁾
 - Arterial Po₂ ≤ 39 mmHg(5.2 kPa)
 - Arterial or venous Pco₂ ≥ 60 mmHg(8.0 kPa) and, **One:**
 - COPD and arterial or venous pH ≤ 7.24
 - No hx of COPD
 - Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- **Episode Day 2, One:** ⁽³³⁾
 - **OBSERVATION, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽³⁴⁾
 - Afebrile
 - Anti-infective, **One:**
 - Arranged
 - Completed
 - Not indicated
 - Infection resolving or ruled out ⁽³⁵⁾
 - Pain controlled or manageable ⁽³⁶⁾
 - Tolerating PO ⁽³⁷⁾
 - No complication or active comorbidity relevant to this episode of care ⁽³⁸⁾
 - **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
 - Acute pain and, **Both:**

- Analgesic 2 to 3 doses within 24h (includes PO) ^(7, 8)
- IV fluid ^(9, 11)
- Infection, actual or suspected and anti-infective ⁽¹³⁾
- **ACUTE, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽³⁴⁾
 - Afebrile
 - Infection resolving or ruled out ⁽³⁵⁾
 - **Anti-infective, One:**
 - Arranged
 - Completed
 - Not indicated
 - Lab values within acceptable limits ⁽³⁹⁾
 - O₂ sat within acceptable limits ^(18, 39)
 - Pain controlled or manageable ⁽³⁶⁾
 - Respiratory impairment improving or resolved
 - Tolerating PO ⁽³⁷⁾
 - No complication or active comorbidity relevant to this episode of care ⁽³⁸⁾
- **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
 - **Aplastic crisis treatment and monitoring, Both:** ⁽¹⁹⁾
 - Hct or Hb monitoring at least 2x/24h
 - IV fluid or blood product transfusion ⁽⁹⁾
 - Pain requiring analgesic ≥ 4x/24h (includes PO) or continuous ^(7, 8)
 - **Respiratory impairment, Both:** ^(15, 32)
 - **Finding, ≥ One:**
 - O₂ sat < 92%(0.92) and < baseline ^(17, 18)
 - **Respiratory symptom unresolved and, ≥ One:**
 - Dyspnea
 - Cough
 - Wheeze
 - Retractions
 - Rales
 - Anti-infective ⁽³²⁾
 - Infection, actual or suspected and anti-infective ⁽¹³⁾
 - Post Critical care ≤ 24h
- **INTERMEDIATE, ≥ One:**
 - NIPPV ⁽²³⁾
 - Oxygen ≥ 40%(0.40) ≤ 2d ⁽²⁴⁾
- **CRITICAL, ≥ One:**
 - ECMO or ECLS ⁽²⁶⁾
 - Exchange transfusion ^(27, 28)
 - Hemodynamic monitoring, invasive ⁽²⁹⁾
 - Mechanical ventilation
 - **NIPPV and, ≥ One:** ⁽²³⁾
 - FiO₂ > 50%(0.50)
 - Respiratory intervention every 1-2h ⁽⁴⁰⁾
 - Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- **Episode Day 3, One:** ⁽⁴¹⁾
 - **OBSERVATION, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽³⁴⁾
 - Afebrile
 - **Anti-infective, One:**
 - Arranged
 - Completed

- Not indicated
- Infection resolving or ruled out ⁽³⁵⁾
- Pain controlled or manageable ⁽³⁶⁾
- Tolerating PO ⁽³⁷⁾
- No complication or active comorbidity relevant to this episode of care ⁽³⁸⁾
- **Non-responder**, not clinically stable for discharge and exceeds episode days at the Observation level of care for this condition, **One**:
 - For sickle cell use Episode Day 3 at a higher level of care
 - For a condition other than sickle cell use appropriate subset (Episode Day 1) based on clinical findings
 - Refer for secondary review
- **ACUTE, One**:
 - **Responder**, discharge expected today if clinically stable last 12h, **All**: ⁽³⁴⁾
 - Afebrile
 - **Anti-infective, One**:
 - Arranged
 - Completed
 - Not indicated
 - Infection resolving or ruled out ⁽³⁵⁾
 - Lab values within acceptable limits ⁽³⁹⁾
 - O₂ sat within acceptable limits ^(18, 39)
 - Pain controlled or manageable ⁽³⁶⁾
 - Respiratory impairment improving or resolved
 - Tolerating PO ⁽³⁷⁾
 - No complication or active comorbidity relevant to this episode of care ⁽³⁸⁾
 - **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One**:
 - Pain requiring analgesic ≥ 4x/24h (includes PO) or continuous ^(7, 8)
 - **Respiratory impairment, Both**: ^(15, 32)
 - **Finding, ≥ One**:
 - O₂ sat < 92%(0.92) and < baseline ^(17, 18)
 - **Respiratory symptom unresolved and, ≥ One**:
 - Dyspnea
 - Cough
 - Wheeze
 - Retractions
 - Rales
 - Anti-infective ⁽³²⁾
 - Infection, actual or suspected and anti-infective ⁽¹³⁾
 - Post Critical care ≤ 24h
- **INTERMEDIATE, ≥ One**:
 - NIPPV ⁽²³⁾
 - Oxygen ≥ 40%(0.40) ≤ 2d ⁽²⁴⁾
- **CRITICAL, ≥ One**:
 - ECMO or ECLS ⁽²⁶⁾
 - Exchange transfusion ^(27, 28)
 - Hemodynamic monitoring, invasive ⁽²⁹⁾
 - Mechanical ventilation
 - **NIPPV and, ≥ One**: ⁽²³⁾
 - FiO₂ > 50%(0.50)
 - Respiratory intervention every 1-2h ⁽⁴⁰⁾
 - Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- **Episode Day 4, One**:
 - **ACUTE, One**:
 - **Responder**, discharge expected today if clinically stable last 12h, **All**: ⁽³⁴⁾

- Afebrile
- **Anti-infective, One:**
 - Arranged
 - Completed
 - Not indicated
- Infection resolving or ruled out ⁽³⁵⁾
- Lab values within acceptable limits ⁽³⁹⁾
- O₂ sat within acceptable limits ^(18, 39)
- Pain controlled or manageable ⁽³⁶⁾
- Respiratory impairment improving or resolved
- Tolerating PO ⁽³⁷⁾
- No complication or active comorbidity relevant to this episode of care ⁽³⁸⁾
- **Partial responder**, not clinically stable for discharge and requires continued stay, ≥ **One:**
 - Pain requiring analgesic ≥ 4x/24h (includes PO) or continuous ^(7, 8)
- **Respiratory impairment, Both:** ^(15, 32)
 - **Finding, ≥ One:**
 - O₂ sat < 92%(0.92) and < baseline ^(17, 18)
 - **Respiratory symptom unresolved and, ≥ One:**
 - Dyspnea
 - Cough
 - Wheeze
 - Retractions
 - Rales
 - Anti-infective ⁽³²⁾
 - Infection, actual or suspected and anti-infective ⁽¹³⁾
 - Post Critical care ≤ 24h
- **INTERMEDIATE, ≥ One:**
 - NIPPV ⁽²³⁾
 - Oxygen ≥ 40%(0.40) ≤ 2d ⁽²⁴⁾
- **CRITICAL, ≥ One:**
 - ECMO or ECLS ⁽²⁶⁾
 - Exchange transfusion ^(27, 28)
 - Hemodynamic monitoring, invasive ⁽²⁹⁾
 - Mechanical ventilation
 - **NIPPV and, ≥ One:** ⁽²³⁾
 - FiO₂ > 50%(0.50)
 - Respiratory intervention every 1-2h ⁽⁴⁰⁾
 - Vasopressor, vasodilator, or inotrope, continuous (excludes low or fixed dose for end stage heart failure)
- **Episode Day 5, One:**
 - **ACUTE, One:**
 - **Responder**, discharge expected today if clinically stable last 12h, **All:** ⁽³⁴⁾
 - Afebrile
 - **Anti-infective, One:**
 - Arranged
 - Completed
 - Not indicated
 - Infection resolving or ruled out ⁽³⁵⁾
 - Lab values within acceptable limits ⁽³⁹⁾
 - O₂ sat within acceptable limits ^(18, 39)
 - Pain controlled or manageable ⁽³⁶⁾
 - Respiratory impairment improving or resolved
 - Tolerating PO ⁽³⁷⁾
 - No complication or active comorbidity relevant to this episode of care ⁽³⁸⁾

- **Non-responder**, not clinically stable for discharge and exceeds episode days at the Acute level of care for this condition, **One**:
 - Use **Extended Stay subset**
 - Refer for secondary review
- **INTERMEDIATE, One**:
 - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Intermediate level of care for this condition, **One**:
 - Use **Extended Stay subset**
 - Refer for secondary review
- **CRITICAL, One**:
 - **Non-responder**, not clinically stable for discharge and exceeds episode days at the Critical level of care for this condition, **One**:
 - Use **Extended Stay subset**
 - Refer for secondary review

Discharge, Level of Care, **One**: ⁽⁴²⁾

- **HOME, Both**:
 - Level of care appropriateness, **Both**:
 - Home environment safe and accessible ⁽⁴³⁾
 - Patient or caregiver demonstrates ability to manage care
 - Sickle cell discharge planning, **All**:
 - Provide comprehensive written discharge and teaching instructions ⁽⁴⁴⁾
 - Education on the importance of, **All**:
 - Vaccination compliance ⁽⁴⁵⁾
 - Monitoring for complications
 - Notifying physician of any change in respiratory status, pain, or fever
 - Medication reconciliation ⁽⁴⁶⁾
 - Follow-up care planned ⁽⁴⁷⁾
 - Identify and address transportation needs
- **HOME CARE, Both**:
 - Level of care appropriateness, **All**:
 - Home environment safe and accessible ⁽⁴³⁾
 - Patient or caregiver willing and able to learn care
 - Outpatient management contraindicated or unavailable ^(48, 49)
 - Skilled services, ≥ **One**: ⁽⁵⁰⁾
 - Skilled nursing ⁽⁵¹⁾
 - Skilled therapy (PT, OT, or ST)
 - Behavioral health ⁽⁵²⁾
 - Sickle cell discharge planning, **All**:
 - Provide comprehensive written discharge and teaching instructions ⁽⁴⁴⁾
 - Education on the importance of, **All**:
 - Vaccination compliance ⁽⁴⁵⁾
 - Monitoring for complications
 - Notifying physician of any change in respiratory status, pain, or fever
 - Medication reconciliation ⁽⁴⁶⁾
 - Follow-up care planned ⁽⁴⁷⁾
 - Identify and address transportation needs
- **BEHAVIORAL HEALTH, Both**:
 - Level of care appropriateness, **One**:
 - Support systems available ⁽⁵³⁾
 - Skilled treatment, ≥ **One**:
 - Clinical assessment ⁽⁵⁴⁾
 - Individual, group, or family counseling

- Psychiatric, substance, or medication evaluation ⁽⁵⁵⁾

● **SKILLED MEDICAL OR THERAPY, Both:**

● Level of care appropriateness, **All:**

- Medical practitioner, NP, PA assessment or oversight ≥ 1x/wk
- Treatment precluded in a lower level

● Services, ≥ **One:**

- Medical and skilled nursing interventions or assessment 1-2x/24h ⁽⁵¹⁾

● Therapy, **All:**

- Goal directed therapy with rehabilitation potential ^(56, 57)
- Functional impairments requiring at least supervision ⁽⁵⁸⁾
- Able to tolerate 1h/d of skilled therapy ≥ 5d/wk

● Complete prior to facility transfer, **All:**

- Medication reconciliation ⁽⁴⁶⁾
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

● **SUBACUTE MEDICAL OR THERAPY, Both:**

● Level of care appropriateness, **All:**

- Medical practitioner, NP, PA assessment or oversight ≥ 2x/wk
- Treatment precluded in a lower level

● Services, ≥ **One:**

- Medical and skilled nursing interventions or assessment 4h/24h ⁽⁵¹⁾

● Therapy, **All:**

- Goal directed therapy with rehabilitation potential ^(56, 57)
- ≥ 2 functional impairments requiring at least minimum assistance ⁽⁵⁸⁾
- Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 1 therapy discipline

● Complete prior to facility transfer, **All:**

- Medication reconciliation ⁽⁴⁶⁾
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

● **SUBACUTE REHAB, Both:**

● Level of care appropriateness, **All:**

- Medical practitioner, NP, PA assessment or oversight ≥ 3x/wk
- Treatment precluded in a lower level

● Services, **All:**

- Goal directed therapy with rehabilitation potential ^(56, 57)
- ≥ 2 functional impairments requiring at least minimum assistance ⁽⁵⁸⁾
- Able to tolerate 2-3h/d of skilled therapy ≥ 5d/wk and ≥ 2 therapy disciplines

● Complete prior to facility transfer, **All:**

- Medication reconciliation ⁽⁴⁶⁾
- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

● **ACUTE REHAB, Both:**

● Level of care appropriateness, **Both:**

- Medical practitioner, NP, PA assessment or oversight ≥ 3x/wk

● Services, **All:**

- Goal directed therapy with rehabilitation potential ^(56, 57)
- ≥ 2 functional impairments requiring at least minimum assistance ⁽⁵⁸⁾
- Able to tolerate ≥ 15h/wk of skilled therapy and ≥ 2 therapy disciplines ⁽⁵⁹⁾

● Complete prior to facility transfer, **All:**

- Medication reconciliation ⁽⁴⁶⁾

- Obtain and complete forms for facility
- Obtain discharge summary and transmit to facility and medical practitioner
- Arrange transportation

● **LONG TERM ACUTE CARE, Both:**

- **Level of care appropriateness, Both:**
 - Treatment precluded in a lower level
- **In lieu of acute or continued hospitalization or failed lower level of care, ≥ One:**
 - Primary condition or illness and treatment of active comorbid condition(s) ⁽⁶⁰⁾
 - Ventilator dependent ≥ 6h/d and weaning planned ^(61, 62)
 - Acute and comorbid conditions requiring prolonged hospitalization ⁽⁶³⁾
- **Complete prior to facility transfer, All:**
 - Medication reconciliation ⁽⁴⁶⁾
 - Obtain and complete forms for facility
 - Obtain discharge summary and transmit to facility and medical practitioner
 - Arrange transportation

Utilization Benchmarks

Condition or Procedure	% Pd Obs	LOS (days)	Type
811 RED BLOOD CELL DISORDERS WITH MCC	n/a	3.8	CMS GMLOS
812 RED BLOOD CELL DISORDERS WITHOUT MCC	n/a	2.8	CMS GMLOS
ACUTE CHEST SYNDROME	7%	6	InterQual
VASO-OCCLUSIVE CRISIS	21%	4.9	InterQual

Percent Paid as Observation indicates the final payment status as a percentage for patients hospitalized with the condition. It may not correspond to the status assigned at admission. Note that patients initially placed in Observation status who are converted to inpatient status (common) are included in the inpatient category, and that patients initially admitted to inpatient status who are converted to observation status (uncommon) are included in the Observation category.

Notes:**1:**

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital-level services **AND** that the patient has a hospital stay of at least two midnights. Short-stay exceptions to this rule are defined below. Application of InterQual® criteria can help an organization comply with the medical necessity requirements of the two-midnight rule.

For those who are impacted by the CMS Two-Midnight Rule, follow the steps below:

1. **Determine the need for hospital-based services by applying criteria.** If a patient meets criteria at any level of care, they have met the medical necessity requirement of the rule.
2. **Determine whether the patient is an inpatient or assign observation status** depending on which level of care is met and the expected length of stay. Based on the presentation of the patient, evidence-based criteria such as InterQual® can support the provider's expectation for length of stay.
 - **When care is expected to span two midnights** and the patient has met criteria at the Acute, Intermediate or Critical level of care, an inpatient status may be appropriate.

NOTE: In the event of an "Unforeseen Circumstance" per CMS resulting in a shorter stay than initially expected, a status of inpatient may still be appropriate:

- Leaving against medical advice (AMA)
- Unexpected death
- Transfer of the patient
- Unexpected clinical improvement
- Election of hospice care

- **When care is expected to span less than two midnights** but Acute, Intermediate, or Critical criteria are met, it may be appropriate to initially assign a status of observation.

NOTE: CMS has defined the following exceptions as being appropriate for inpatient status regardless of length of stay:

- Mechanical Ventilation, established during current hospitalization
- Surgical procedure or intervention found on the CMS Inpatient Only List

- **When it is undecided as to whether care will span two midnights**, the reviewer may use the level of care met as a guide to determine whether the patient is appropriate as an inpatient or observation status. The criteria consider multiple factors, including clinical severity, comorbidity, and risk, when differentiating between the Observation and Acute levels of care. These factors impact the length of stay. A patient meeting criteria at the Acute levels of care can generally be expected to have a length of stay of greater than two midnights and thus could be assigned to inpatient status. Patients who meet criteria at the Observation level of care are more appropriate for observation status.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

2:

Initial Review criteria are a level of care determination tool, intended to be used as **real-time** decision support in the emergency department to identify if observation or inpatient hospital level services are warranted. They help the reviewer determine whether a patient is appropriate for observation or inpatient admission at the time a decision to admit the patient is being made. Initial Review criteria evaluate **only data that are available at the time the decision is being made**. This may include previously provided interventions or the results of laboratory, imaging, and other tests. Initial Review may be appropriate when bridge or holding orders (e.g., admit to telemetry) are in place. These orders are intended to address the patient's needs until full treatment and medication orders are written.

Initial Review criteria **should not be used** retrospectively or once the decision to admit has been made (e.g., when full treatment and medication orders have been written). If the reviewer has sufficient information to complete an Episode Day 1 review, an Initial Review need not be completed.

Note: While Initial Review enables identification of the level of care, it is not intended to, and should not be used as a substitute for an Episode Day 1 review, which will generally include specific, evidence-based interventions and

intensity of service requirements.

For further clarification, see the Acute Level of Care Review Process.

3:

Hemodynamically stable patients who require at least 6 hours, and for certain conditions, up to 48 hours, of treatment or assessment pending a decision regarding the need for additional care. Observation level of care services may be provided in designated observation units or on a hospital floor.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions consider a variety of factors, such as clinical judgment, medical necessity, payer policies and contracts, and state and national regulations.

4:

To reduce the risk of developing acute chest syndrome in patients with vaso-occlusive crisis, incentive spirometry is recommended (Ballas, Clinical hemorheology and microcirculation 2018, 68: 105-28; Jain et al., Pediatric allergy, immunology, and pulmonology 2017, 30: 191-201).

5:

These criteria refer to the lack of response to treatment received through the medical practitioner's office, outpatient clinic (e.g., urgent care), teleconsultation, trial of home care services, emergency department (prior to decision to admit), or hospital Observation setting (if appropriate for this condition). These criteria can be applied when the patient fails to respond to more intensive treatment beyond the routine treatments or medications delivered for a chronic condition.

6:

The routes for outpatient analgesics include:

- Oral (PO)
- Subcutaneous (SC)
- Intranasal
- Transdermal

7:

In the setting of sickle cell crisis and severe pain, rapid initiation of parenteral opioids within 60 minutes of presentation is recommended, followed by a scheduled administration regimen. Patient controlled analgesia (PCA) should be considered due to the following advantages (Brandow et al., Blood advances 2020, 4: 2656-701):

- Minimizes fluctuation of serum drug levels
- Reduces the time between the onset of pain and the administration of the analgesic
- Prevents excessive sedation

Opioid selection and dosing should be individualized based on history of use and home regimen. Due to the risk of neurotoxicity, meperidine is no longer recommended and should be reserved for patients who fail to respond to alternative agents. Adjunctive pharmacologic therapy may include non-steroidal anti-inflammatory drugs and antihistamines to relieve itching. Although providers should be cautious when considering anti-inflammatory medications because of the risk of renal toxicity in the sickle cell population. The decision to continue long-acting oral opioids used to manage chronic pain should be made on a case-by-case basis. In most circumstances, chronic pain regimens should be continued to ensure adequate pain relief while preventing withdrawal. Close monitoring of oxygenation and sedation levels is recommended to prevent over-sedation (Brandow et al., Blood advances 2020, 4: 2656-701).

8:

Evidence is limited on the frequency of analgesic medication administration at the Observation level of care. Many sickle cell patients are on prescribed opioids at home and require higher doses than other patient populations during an acute pain crisis. The American College of Emergency Physicians developed a sickle cell pathway in September 2020 which has yet to be widely implemented. This pathway would allow for sickle cell patients on patient controlled analgesia (PCA) to be at the Observation level of care for up to 24 hours. Currently, evidence

suggests that sickle cell patients who are receiving analgesic medication ≥ 4 times in 24 hours or who are on continuous analgesia (including PCA) be treated at the Acute level of care (Brandow et al., Blood advances 2020, 4: 2656-701; Lyon et al., Annals of emergency medicine 2020, 76: S12-S20).

9:

Instruction: This line of criteria is intended for the patient receiving condition-directed maintenance IV fluid and excludes the patient who has IV fluid ordered to keep vein open (KVO).

10:

Vaso-occlusive crisis requiring fluid administration should not be infused at a rate higher than normal maintenance. Fluid may be administered as a combination of intravenous and oral to achieve the desired rate. Caution should be used as excessive fluid intake may precipitate or exacerbate acute chest syndrome (Brandow et al., Blood advances 2020, 4: 2656-701; Ballas, Clinical hemorheology and microcirculation 2018, 68: 105-28).

11:

There is limited evidence on the efficacy of pain relief with the administration of fluids in addition to pharmacological pain management during an acute pain crisis. However, fluid is frequently administered to these patients as a means of reestablishing euvolemia. Caution must be taken when administering fluid especially to adult sickle cell patients due to potential exacerbation of acute chest syndrome and cardiac risks (Brandow et al., Blood advances 2020, 4: 2656-701).

12:

The Centers for Disease Control and Prevention (CDC) define a fever as greater than or equal to 100.4 degrees Fahrenheit (i.e., greater than or equal to 38 degrees Celsius) and do not specify the route of measurement (Centers for Disease Control and Prevention, Definitions of symptoms for reportable illness. 2017). The route utilized may be directed by institutional protocols, equipment availability, patient preference, or other patient-specific factors.

13:

Sickle cell patients presenting with fever of 100.4°F (38.0°C) - 101.2°F (38.4°C) in which infection is suspected but the source is unknown are to be admitted to the Observation level of care for a thorough workup. The most common tests in a workup for these patients include blood and urine cultures, complete blood count, and a chest radiograph to rule out pneumonia and acute chest syndrome. For a fever of $\geq 101.3^\circ\text{F}$ (38.5°C), sickle cell patients are admitted to the Acute level of care for closer monitoring, anti-infective initiation and a more thorough workup (van Tuijn et al., American Journal of Hematology 2020;; National Heart Lung and Blood Institute, Evidence-based management of sickle cell disease: Expert panel report. 2014; Yawn et al., JAMA 2014, 312: 1033-48).

14:

Hemodynamically stable patients who require treatment, assessment, or intervention every 4 to 8 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

15:

Acute chest syndrome is diagnosed by the presence of a new infiltrate on chest radiograph accompanied by either chest pain, an acute oxygen desaturation, an elevated temperature or a new respiratory symptom (Chou et al., Blood advances 2020, 4: 327-55; van Tuijn et al., American Journal of Hematology 2020;; Ballas and Darbari, Complementary Therapies in Medicine 2020, 49: 102327).

16:

Presenting symptoms of acute chest syndrome resemble pneumonia and may include fever, cough, chest pain, or increased work of breathing as evidenced by dyspnea, wheezing, retractions, and rales. Although a new pulmonary infiltrate on chest radiograph is considered a hallmark sign of acute chest syndrome, results may appear normal or non-diagnostic in the early stages. Pleural effusions, multilobe disease, and persistent hypoxia despite supplemental oxygen therapy are predictive of a severe course (Liem et al., Blood advances 2019, 3: 3867-97; Ballas, Clinical

hemorheology and microcirculation 2018, 68: 105-28).

17:

Baseline refers to either the patient's normal baseline or a newly established baseline. In the absence of documentation, a patient's baseline status may be presumed to be normal.

18:

Supplemental oxygen is recommended when the oxygen saturation in a patient with acute chest syndrome is < 92%. If the patient's known baseline is below 92%, supplemental oxygen is not necessary unless the oxygen saturation drops below known baseline. Caution should be used when administering oxygen as excessive or unnecessary oxygen delivery may suppress erythrocyte production (Chou et al., Blood advances 2020, 4: 327-55; Lippert, In: Emergency Management of Infectious Diseases. 2018:473).

19:

Aplastic crisis is marked by a rapidly declining hemoglobin level or a reduction of the reticulocyte count. Clinical signs and symptoms may include tachycardia, lethargy, fatigue, dyspnea, and syncope. Aplastic crisis is typically attributed to infection and is most commonly seen in children. Repeated episodes of aplastic crisis are rare due to lifelong humoral immunity that results from the infection. Treatment requires prompt transfusion of red blood cells and is frequently accompanied by the infusion of intravenous fluids (Ballas, Clinical hemorheology and microcirculation 2018, 68: 105-28; Howard, Hematology 2016, 2016: 625-31; National Heart Lung and Blood Institute, Evidence-based management of sickle cell disease: Expert panel report. 2014).

20:

If a patient's baseline is known, aplastic crisis is diagnosed when patient's hemoglobin level is 2 g/dL less than their baseline. If baseline is unknown, aplastic crisis is diagnosed when a sickle cell patient's hemoglobin is less than 5.5 g/dL. These patients require either blood product transfusion or IV fluid infusion and frequent lab monitoring until consistently stable or returned to baseline (Ballas, Clinical hemorheology and microcirculation 2018, 68: 105-28; Saunthararajah and Vichinsky, In: Hematology. 2018:584-607. e5).

21:

Continuous refers to the administration of a medication via a continuous infusion. Examples of continuous analgesic infusions include basal rate infusions and patient controlled analgesia.

22:

Hemodynamically stable patients who require treatment, assessment, or intervention every 2 to 4 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

23:

Noninvasive positive pressure ventilation (NIPPV), also known as NIV, provides respiratory support by application of a tightly fitting facial mask, nasal mask, or helmet rather than an endotracheal tube or tracheostomy. In some cases, the use of NIPPV can avoid the need for endotracheal intubation and decreases the risk of barotrauma, lung injury, and/or infection. NIPPV is commonly delivered by a bilevel positive airway pressure ventilator (BiPAP), a continuous positive airway pressure device (CPAP), or a mechanical ventilator. Supplemental oxygen can be delivered at concentrations approximating 100%(1.0).

The decision to provide respiratory support via NIPPV, and the modality to provide NIPPV, is based upon the patient's specific clinical findings (e.g., medical condition leading to respiratory failure, underlying comorbidities, clinical progression, improvement). Examples of NIPPV devices are volumetric (i.e., deliver a defined volume), barometric (i.e., deliver a defined pressure), and combined (i.e., deliver defined volumes and pressure). The terminology for NIPPV delivery systems may differ between equipment manufacturers and provider organizations. InterQual® criteria do not differentiate between the different NIPPV modalities.

Whether or not high-flow nasal cannula (HFNC) should be classified as a component of NIPPV is unclear, and there is conflicting evidence in the medical literature. Where HFNC is an appropriate modality, InterQual® defines this in a specific criteria point. As such, NIPPV does not include HFNC (Hackett, A., PulmCCM 2018; Nardi et al., F1000Res

2017, 6: 290; Osadnik et al., Cochrane Database Syst Rev 2017, 7: CD004104; Allison and Winters, Emerg Med Clin North Am 2016, 34: 51-62; Gregoretti et al., Crit Care Clin 2015, 31: 435-57).

24:

Oxygen therapy is the administration of oxygen at concentrations greater than ambient air (room air: 21%(0.21)) with the intent of treating and/or preventing the symptoms and manifestations of hypoxia. The oxygen concentration or percentage (FiO₂) delivered varies with the manufacturer's design, oxygen flow rate, the patient's respiratory rate, and tidal volume. Actual inspired FiO₂ with nasal cannula varies significantly with the patient's minute ventilation and pattern of breathing. For patients at risk for CO₂ retention, where a precise inspired FiO₂ is required, the Venturi mask is the preferred method. In general, patients who are very ill or have respiratory disease may require considerably higher flow rates to achieve the desired FiO₂. The following are estimates of O₂ delivered at the associated flow rates:

LOW-FLOW SYSTEMS**Nasal cannula**

Room air 21%(0.21)

1L 24%(0.24) **4L** 36%(0.36)

2L 28%(0.28) **5L** 40%(0.40)

3L 32%(0.32) **6L** 44%(0.44)

Simple oxygen masks

- 35-50%(0.35-0.50) FiO₂ (5-10 L/min)
- Flow rates are usually maintained at 5 L/min or more to avoid accumulation of CO₂ in the mask

Venturi masks

- 24-50%(0.24-0.50) FiO₂ (4-12 L/min)

Partial rebreathing masks

- 40-70%(0.40-0.70) FiO₂ (6-10 L/min)

Non-rebreathing masks

- 60-80%(0.60-0.80) FiO₂ (minimum flow of 10 L/min)

HIGH-FLOW SYSTEMS**Air-entrainment masks/nebulizers**

- 24-40%(0.24-0.40) FiO₂

Heated, humidified high-flow concentrating nasal cannula (HFNC)

- 24-100%(0.24-1.0) FiO₂

25:

Hemodynamically unstable patients (or those with the potential to become unstable) who require treatment, assessment, or intervention every 1 to 2 hours.

Note: InterQual® Level of Care criteria are intended to be used to support a patient-specific clinical determination of medical necessity and not for final billing or reimbursement decisions. Final payment decisions are based on a variety of factors including medical necessity, payer rules, and contracts.

26:

Extracorporeal membrane oxygenation (ECMO) or extracorporeal life support (ECLS) provides oxygen and removes carbon dioxide from blood. ECMO/ECLS facilitates the recovery of cardiopulmonary function when other measures fail to improve gas exchange. Indications for the use of ECMO/ECLS in children include refractory septic or cardiogenic shock, cardiac arrest as a bridge to transplant, or refractory hypoxemic or hypercarbic respiratory failure. There is clear evidence to support the use of ECMO/ECLS in the pediatric population, but definitive studies in adults are lacking. However, in adults, observational studies and other data support the use of ECMO/ECLS for severe refractory, hypoxemic, or hypercarbic respiratory failure (Mehta et al., American College of Cardiology Perspectives and Analysis 2018; Napp et al., Herz 2017, 42: 27-44; Guirand et al., J Trauma Acute Care Surg 2014, 76: 1275-81). ECMO/ECLS incorporates the use of an external circuit, a cannula for vascular access, a pump to propel the blood through the circuit, and an artificial membrane that oxygenates the blood. Common complications include hemorrhage, hemolysis, and infection. Stroke, embolism, and seizures in children have also been reported.

Infants with intraventricular hemorrhage (IVH) or those at high risk for IVH (i.e., weight less than 2000 grams, age less than 35 weeks) are not candidates for this therapy (Abrams et al., J Am Coll Cardiol 2014, 63: 2769-78; Domico et al., Pediatr Crit Care Med 2012, 13: 16-21; MacLaren et al., Intensive Care Med 2012, 38: 210-20; Peek et al., Health Technol Assess 2010, 14: 1-46; Mugford et al., Cochrane Database Syst Rev 2008: CD001340).

27:

An exchange transfusion is the removal of blood or blood components and replacement with donated blood products or IV fluids, usually normal saline. There are three main types of exchange transfusions: double volume exchange for severe hyperbilirubinemia, partial exchange transfusion for polycythemia or hyperviscosity, and red cell exchange for severe sickle cell crisis. Possible side effects may include hypotension, hypocalcemia, infection, emboli, thromboses, and death.

28:

In the setting of sickle cell crisis, an exchange transfusion may be considered in patients with a hemoglobin of 10 g/dL (100 g/L) or greater and a hemoglobin S level of 30%(0.30) or greater. It may also be helpful for patients with severe or worsening symptoms such as multilobe involvement, persistent or worsening hypoxemia, neurologic abnormalities, or multisystem organ failure (Chou et al., Blood advances 2020, 4: 327-55; Ballas, Clinical hemorheology and microcirculation 2018, 68: 105-28).

29:

Invasive hemodynamic monitoring may include at least one of the following methods of assessment: arterial line, pulmonary artery catheter, central venous catheter, or cardiac output monitoring (Rajaram et al., Cochrane Database Syst Rev 2013, 2: CD003408).

30:

Transition Plan: InterQual's® Transition Plan identifies patients at high risk for readmission who may benefit from a comprehensive discharge plan. Sickle cell patients are at risk for readmission (Brousseau et al., JAMA 2010, 303: 1288-94).

31:

This subset is for the review of patients with an infection in which the source has not yet been identified. This includes criteria for patients with a fever and infectious symptoms which have failed outpatient management. Once a source is identified, patients are to be directed to the appropriate infection specific subset.

32:

For patients with acute chest syndrome, parenteral antibiotic coverage is recommended. Other treatments can include supplemental oxygen, corticosteroids, bronchodilators, and nitric oxide (Ballas, Clinical hemorheology and microcirculation 2018, 68: 105-28; Chou et al., Blood advances 2020, 4: 327-55).

33:

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital level services **AND** that the patient has a hospital stay of at least two midnights.

For patients who are in observation status and do not meet Observation responder criteria on Episode Day 2, apply Episode Day 2 criteria at the Observation, Acute, Intermediate or Critical level of care in the appropriate subset to demonstrate the continued need for hospital-based services.

NOTE: For review of a surgical patient, Post-Operative Day 1 equates to Episode Day 2.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

34:

Selection of this criteria point indicates that the patient is responding to treatment and is clinically stable for transfer or discharge. To determine the most appropriate post-acute level of care, see discharge criteria.

35:

Resolution of infection is indicated by a decrease in temperature, a normalizing of white blood cells and other lab values, and improving imaging studies. If there is a wound present, signs of infection should be resolving (e.g., decrease in wound drainage, less pain at site).

36:

Pain is considered to be adequately controlled when a stable medication regimen and/or adjunct therapy provides sufficient, symptomatic relief without causing significant adverse side effects. Studies have shown epidural analgesia to be very effective in the initial post-operative period (McNicol et al., Cochrane Database Syst Rev 2015: CD003348; Popping et al., Ann Surg 2014, 259: 1056-67). For acute pain, the Center for Disease Control (CDC) recommends prescribing the lowest effective dose of immediate-release opioids, with careful attention paid to the quantity and duration of the prescription. Three days or less will often work; more than 7 days is rare. For chronic pain unrelated to active cancer, palliative or end-of-life care, the CDC recommends the following guidelines for clinicians (Dowell et al., Jama 2016, 315: 1624-45; Dowell et al., MMWR Recomm Rep 2016, 65: 1-49):

- Consider nonpharmacologic therapy and nonopioid pharmacologic therapy first
- Know the patient's history of controlled substance prescriptions, using the state prescription drug monitoring program (DMP) data when initiating opioid therapy and then every 3 months thereafter on every prescription
- Weigh the risks and benefits of opioid therapy before starting or continuing opioids
- Establish realistic treatment goals, discuss risks and benefits of opioid therapy, and go over the plan for discontinuing opioids if the harm outweighs the benefits
- Discuss patient and clinician responsibilities for managing this therapy. If a patient has an opioid use disorder, offer evidence-based, medication-assisted treatment in combination with behavioral therapies

When prescribing opioids for chronic pain:

- Combine them with nonpharmacologic therapy/nonopioid therapy
- Prescribe immediate-release opioids versus extended-release or long-acting opioids
- Start at the lowest dose
- Evaluate the benefits and the harm with patients within 1 to 4 weeks of starting opioid therapy, and then at least every 3 months thereafter
- Consider annual urine testing to assess for prescribed medications as well as other controlled prescription and/or illicit drugs
- Avoid prescribing opioid pain medication and benzodiazepines concurrently

For patients in the early post-surgical phase, a large retrospective cohort study found that the duration of the opioid prescription, rather than its dosage, was more strongly associated with opioid misuse (Brat et al., BMJ 2018, 360: j5790).

37:

Oral intake should be greater than or equal to the patient's maintenance fluid requirements or within parameters which the medical practitioner determines are acceptable. Maintenance fluid requirements should be increased in the presence of dehydration or insensible loss. Maintenance fluid requirements (*mL/kg/d*) can be calculated based on current body weight (*kg*):

- 100 *mL/kg/d* for first 10 *kg*
- 50 *mL/kg/d* for second 10 *kg*
- 20 *mL/kg/d* for each additional *kg* in weight

38:

Relevant complications or comorbidities are conditions that actively require management during an inpatient stay.

39:

When a criteria point states "within acceptable limits," it refers to either the patient's normal baseline, a newly established baseline, or parameters that the medical practitioner determines are acceptable.

40:

Respiratory interventions include ventilator/noninvasive positive pressure ventilation (NIPPV) setting changes, chest physiotherapy, suctioning, and nebulizer or metered-dose inhaler (MDI) treatments.

41:

The CMS Two-Midnight Rule generally requires that the medical record support a patient's need for hospital level services **AND** that the patient has a hospital stay of at least two midnights.

For patients who are in observation status beyond the second midnight (Episode Day 3) and who do not meet Observation responder criteria, apply Episode Day 3 Inpatient criteria in the appropriate condition-specific or general subset.

Note: Patients who meet criteria **at any of** the Inpatient levels of care **based on medically necessary hospital-based services and supported by clinical documentation would be appropriate for inpatient status**; those who do not meet criteria would be appropriate for secondary review.

(Centers for Medicare & Medicaid Services, Title 42; Part 412; Subpart A - Admissions. 2023; Centers for Medicare & Medicaid Services. Title 42; Part 422; Subpart C-Benefits and Beneficiary Protections. 2023; Centers for Medicare & Medicaid Services, In: BFCC-QIO 2 Midnight Claim Review Guideline. 2022)

42:

The discharge screen is a resource tool and not criteria. Referring to the discharge screen at the initiation of discharge planning is recommended.

43:

The home environment and safety assessment will normally include an evaluation of the patient's pre- and post-hospitalization functional level, physical layout of the home (e.g., entrances and exits, stairs, access to the community), identification of unsafe conditions (e.g., scatter rugs, missing handrails, oxygen use and smoking, lack of fire safety devices, inadequate lighting, heating, and cooling), factors that may trigger symptoms (e.g., secondhand smoke, poor food choices, dysfunctional family dynamics), sanitation hazards (e.g., lack of electricity, running water, refrigeration, inadequate toilet facilities, presence of insects or rodents), and the need for adaptive equipment (e.g., walker, handrails for the tub or shower, elevated toilet seat).

44:

Ensuring the patient and/or caregiver understands all aspects of the condition and can assume responsibility for self-care is crucial in preventing readmission. Assessing a patient and/or caregiver's level of understanding after education has been provided can be difficult. The teach-back method is an effective way of providing education at the appropriate level and can also be used to assess the learner's comprehension. To use the teach-back method, discuss key points in common terms and avoid using medical jargon or unfamiliar terms. After the education has been delivered, ask the learner to repeat what was learned. Gaps or misinterpretations in the learner's explanation will pinpoint areas where communication may have failed and provide opportunity for clarification.

45:

Patients with sickle cell disease should follow the routine vaccination schedule recommended by the Advisory Committee on Immunization Practices, including coverage against pneumococcal infections, meningococcal infections, and *Haemophilus influenzae* type B disease (Ezeanolue, E., et al. General Best Practice Guidelines for Immunization. Best Practices Guidance of the Advisory Committee on Immunization Practices (ACIP). 2017; National Heart Lung and Blood Institute, Evidence-based management of sickle cell disease: Expert panel report. 2014).

46:

Medication reconciliation is a formal process or technique used by health care providers and pharmacists to identify the most complete and accurate list of all medications a patient is taking at times of transitions in care (e.g., upon hospital admission, transfer from one unit to another during hospitalization, or discharge from the hospital to home or another facility). The goals of this process are to ensure medication and dosages are appropriate for the patient, resolve discrepancies in drug regimens, and ultimately prevent medication errors and reduce adverse drug events. Medication reconciliation is a Joint Commission National Patient Safety goal. Coordinating information when a patient is transferred to another setting, service, practitioner, or level of care ensures accurate medications are listed. The process is comprised of the following (Hospital: National Patient Safety Goals. 2020):

- Obtain an external list of medications (e.g., medications taken prior to admission)
- Develop a list of current medications and add them to the medical record
- Compare the medications from the external list to the current list
- Clarify inconsistencies/discrepancies (e.g., omissions, duplications, contraindications, unclear information, and changes)
- Develop a list of medications to be prescribed at discharge or transfer
- Communicate the new list to the patient and/or appropriate caregiver(s)
- Ensure that patient and/or caregiver(s) understand the medication information upon discharge

Specific medication issues identified as being problematic include missing medication information from transfer orders, lack of information on medications provided in the acute setting, incomplete medication records, discrepancies between hospital regimen and discharge summary, and missing information pertaining to the patient's tolerance of a medication regimen. Although medication reconciliation is an important aspect in patient safety, there is a lack of consensus and evidence about the best effective methods of implementing this process. Physician-led and electronic medication reconciliation in hospitals are effective strategies to reduce medication discrepancies, however, the impact of these interventions is uncertain due to the low quality of evidence (Choi and Kim, J Clin Pharm Ther 2019, 44: 932-45; Redmond et al., Cochrane Database Syst Rev 2018, 8: CD010791). Trained pharmacy technicians, under the direction of a licensed pharmacist, may be an option for developing and expanding medication reconciliation processes (Irwin et al., Hosp Pharm 2017, 52: 44-53).

47:

Close follow-up of older adults, pediatric, or at-risk patients after discharge can help minimize hospital readmission and total health care costs (Rising et al., Ann Emerg Med 2013, 62: 145-50; Frei-Jones et al., Pediatr Blood Cancer 2009; 52(4): 481-485; Kripalani et al., JAMA 2007; 297(8): 831-841). Studies have shown that for greater than 50% (0.50) of patients re-hospitalized within 30 days of discharge, there was no visit to a physician's office between the time of discharge and re-hospitalization (Hernandez et al., JAMA 2010, 303: 1716-22; Jencks et al., N Engl J Med 2009; 360(14): 1418-1428). Efforts to reduce readmissions should focus on patients known to be at high risk and patients who have frequent visits to the emergency department (ED). Patients at high risk may have multiple ED visits prior to readmission (Rising et al., Ann Emerg Med 2013, 62: 145-50).

48:

Outpatient management contraindicated refers to the following:

- The patient whose medical condition is so fragile that going to the clinic or medical practitioner's office for treatment would put the patient at risk
- The patient with extensive equipment (e.g., ventilators, specialized beds) where travel outside the home would be cumbersome and places the patient at risk
- The treatment goal is to adapt the home environment for increased independence (e.g., an individual who is blind who needs mobility education)

49:

Outpatient management unavailable refers to the following:

- The patient whose treatment (e.g., dressing changes) is so extensive that it takes too long or is too frequent (e.g., twice daily) to be done in an outpatient clinic or medical practitioner's office
- The scheduling of the treatment falls outside the medical practitioner's office or clinic hours
- Travel distance to receive care is excessive (e.g., greater than 1 hour)
- The service is unavailable in the outpatient setting

50:

Skilled nursing services refer to services that must be provided by a licensed nurse qualified to assess and monitor patient condition(s) and provide medical treatment and/or teaching to patients who have skilled nursing needs. Examples of Home Care level of care interventions include skilled assessment, intervention, or education for any of the following:

- New treatment or medication regimen
- Adjustment in the treatment or medication regimen
- Infusion therapy administration, teaching, or access device management
- Management and evaluation of a care plan for patients with an active comorbidity and multiple care needs

- Chronic disease requiring a disease management program (e.g., asthma, heart failure, chronic obstructive pulmonary disease, diabetes)
- End-stage disease, hospice, or palliative care

51:

Skilled nursing interventions refer to services that are provided or supervised by a qualified or licensed professional when the care needs of the patient require assessment, observation, monitoring, and/or education to achieve the medically desired result. Close observation and assessment by nursing may be necessary at this level of care. This includes nursing care such as medication administration (excludes routine), vital signs monitoring, and/or other interventions not listed.

52:

The Home Care criteria are used for a patient who has been admitted or is expected to be admitted for home care. When criteria are met for this level of care and it is not available in your area, we recommend you use the next higher level of care.

Home care can include specialty services provided by a psychiatric nurse following a specialized treatment plan developed by a psychiatrist or advanced practice registered nurse. It is intended for treatment and monitoring of a patient's psychiatric condition that prevents the patient from leaving the house or makes leaving the house extremely difficult or unsafe (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022). Home care can also be used for a patient with severe and persistent mental illness with a history of nonadherence to medication.

Evaluation and Treatment

Programming may differ based upon legislative and geographical variances and is subject to organizational policy; however, at a minimum, it should include:

- Care coordination with treating physician and other providers
- Clinical assessment every visit
- Education every visit
- Medication assessment, teaching, and management every visit
- Medication reconciliation initiated within first visit
- Psychosocial assessment within first visit
- Substance evaluation within first 2 visits

For those patients who are covered under a Medicare benefit, the Affordable Care Act mandates that a certifying physician or allowed practitioner have a face-to-face visit with the patient within 90 days prior to the start of care, or within 30 days after the start of care. There must be documentation of the patient's condition that supports the need for home care services and the requirement of homebound status (Centers for Medicare & Medicaid Services. Chapter 7, In: Medicare benefit policy manual. 2022).

53:

Support system includes friends, family, social services, health care delivery providers or agencies, sponsors, clergy, and self-help groups.

54:

The initial clinical assessment is performed by a licensed professional and includes a comprehensive review of the patient's presenting diagnosis and a review of body systems. Identification of current and potential medical needs and health problems can be identified by the clinical assessment. The clinical assessment includes:

- History and physical exam (e.g., vital signs, height, weight)
- Pain assessment includes cause, intensity, quality, onset, duration, and effects on quality of life (e.g., activities of daily living (ADLs), instrumental activities of daily living (IADLs), and interpersonal relationships). Pain relievers should also be included
- Nutritional and hydration status
- Functional ability
- Safety and infection control measures
- Prescribed and over-the-counter (OTC) medications, herbal supplements, and home remedies
- Patient's and/or caregiver's understanding of medication use, dosing, side effects, and adherence to the medication or treatment regimen
- Patient's and/or caregiver's understanding of the illness or disease process and potential long-term

complications

- Patient's and/or caregiver's coping strategies to deal with the illness or injury and ability to follow the plan of care

55:

For specific level of care recommendations for the treatment of psychiatric and substance use disorders, refer to the InterQual® Adult and Geriatric Psychiatry, InterQual® Child and Adolescent Psychiatry, or InterQual® Substance Use Disorders products.

56:

Goal-directed therapy includes one or more of the following:

- Activities of daily living (ADLs) (e.g., feeding, dressing, bathing, grooming, toileting, functional mobility)
- Bed mobility or transfers (e.g., transfers to chair, toilet, commode, tub, shower, car)
- Cognitive, speech, language, or swallowing retraining
- Home and lifestyle modification (e.g., assessment and training focused on the home, job, school, or work environment)
- Pain management techniques (e.g., assessment and interventions used to relieve pain or decrease its intensity)
- Pulmonary rehabilitation (e.g., energy conservation, speech, breathing re-education, postural drainage, percussion, or vibration)
- Positioning and splinting
- Range of motion and strengthening
- Wheelchair mobility
- Ambulation

57:

Rehabilitation potential refers to the probability that therapy and medical goals are realistic and attainable based on patient's prior level of function, severity of illness or injury, and the extent of impairments.

58:

Functional assistance levels are based upon the patient's function during tasks and activities necessary to return to household mobility or ambulation. The functional assistance level required for each individual task or activity (e.g., mobility, activities of daily living (ADLs)) may vary.

The following terms are commonly used in the post-acute setting:

- Independent - Patient can safely and within a reasonable amount of time perform a task (or developmentally appropriate task) without physical or cognitive assistance or supervision
- Modified Independent - Patient performs an activity with a supportive device, adaptive equipment, and/or prosthetic or orthotic device. Additional time may be required to complete the activity and/or there are safety (risk) considerations
- Supervision - Patient performs an activity with standby or distant supervision or setup. Verbal cueing or coaxing, without physical contact or setup of items and application of orthoses may be required when patient's safety awareness is impaired
- Minimum or limited Assistance - Patient performs at least 75%(0.75) of an activity and requires some physical contact to steady, guide, or move
- Moderate or extensive Assistance - Patient performs at least 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Maximum Assistance - Patient performs 25%(0.25) to 50%(0.50) of an activity and requires physical assistance for functional mobility or ADLs
- Total Assistance or dependence - Patient performs less than 25%(0.25) of an activity and may require total assistance for functional mobility or ADLs

59:

The intensity of rehabilitation therapies is a driver for admission to an acute inpatient rehabilitation facility. Documentation should reflect the need for required and ongoing intensive therapies. The determination of whether a patient requires and can tolerate an intensive rehabilitation program of at least 15 hours is not based on any one single piece of documentation. All clinical documentation should be evaluated by the reviewer to determine

whether the patient is demonstrating the need for services and has the ability to tolerate them. Patients who do not require this level of intensive therapy should be treated at a lower level of care. Rehabilitation therapies should be reasonable and medically appropriate and should not exceed a patient's level of tolerance. If there are clinical features which preclude the patient's ability to tolerate the intensity threshold of at least 15 hours per week, secondary review would be appropriate (Centers for Medicare & Medicaid Services, Medicare Benefit Policy Manual Chapter 1 - Inpatient Hospital Services Covered Under Part A. 2021; Miller and Forer, Pm r 2021, 13: 535-9; Forrest et al., Medicine (Baltimore) 2019, 98: e17096).

60:

Comorbid conditions include new onset or worsening behavioral symptoms, heart failure, deep vein thrombosis, uncontrolled diabetes, immunocompromised states, malignant or end-stage disease, malnutrition, renal insufficiency or end-stage renal disease, infection, ventilator dependence, noninvasive positive pressure ventilation or respiratory insufficiency, complex wound care, and new functional impairment(s) with limitation.

61:

Ventilation includes invasive ventilation and noninvasive positive pressure ventilation (NIPPV), including continuous positive airway pressure (CPAP) and bilevel positive airway pressure (BiPAP).

62:

Notable causes for prolonged mechanical ventilation include:

- Ventilator-acquired pneumonia
- Renal failure
- Heart failure
- Exacerbation of chronic obstructive pulmonary disease (COPD)
- Chest wall disorder
- Neuromuscular diseases
- Traumatic brain injury
- Sepsis
- Complications following cardiac surgical procedures

Major contributing factors for failure to wean include (Ambrosino and Vitacca, Multidiscip Respir Med 2018, 13: 6; Davies et al., Br J Anaesth 2017, 118: 563-9):

- Increased work of breathing
- Impaired respiratory drive
- Inspiratory muscle weakness

63:

Long-Term Acute Care (LTAC) is a recognized level of care designation by the Centers for Medicare and Medicaid Services (CMS) for acute care hospitals. It is defined by federal statutes as a level of care for patients requiring an average length of stay of 25 days or more (Centers for and Medicaid Services, Fed Regist 2018, 83: 41144-784). Patients who require a short stay for treatment may be more appropriate for the subacute or skilled nursing facility (SNF) level of care.